

SOLE SOURCE JUSTIFICATION FOR ACQUISITION UNDER THE SIMPLIFIED ACQUISITION THRESHOLD

PR: 0040666366

1. Contracting Agency and Activity.

The Department of the Interior, United States Geological Survey, Geology, Geochemistry and Geophysics Science Center, plans to contract on a sole source basis. This document sets forth the justification and approval as required by FAR 13.106-1(b)(1).

2. Nature of Action Being Approved.

Purchase of an upgraded Laser Induced Breakdown Spectrometer (LIBS) system (Tandem Prism Upgrade) used for acquiring elemental compositional data associated with laser ablation inductively coupled plasma analysis (LA-ICP-MS).

3. Description of Supplies or Services.

This purchase is for an upgraded Tandem Prism LIBS detection system that is customized designed for the current Resolution SE (RAI 1025) laser system currently housed in the PlasmaLab Room 2432, Building 95, Denver Federal Center. The LIBS system includes a new, higher sensitivity spectrometer, computer control system, control software (Clarity NeXt) that is fully integrated into the existing laser system software (Geostar), and a custom ablation cell window to allow optimum LIBS signal to the detector. This purchase also includes a second laser cell for use with ultra clean samples and conditions. Awardee: Applied Spectra.

4. Estimated Dollar Value.

The cost of the LIBS spectrometer, software, ablation cell window and extra ablation cell is \$ [REDACTED] and this includes shipping, training, installation, etc.

5. Statutory Authority.

The proposed action may be awarded on a sole source basis under the authority of FAR 13.106-1(b)(1) – only one source reasonably available.

6. Rationale Supporting Use of Citation in No. 5.

Applied Spectra is the company that built the existing laser ablation system (Resolution SE, RAI 1025) valued at \$ [REDACTED] and offers the LIBS detector as an upgrade to the analytical capabilities of this current system. Because of the simultaneous nature of the data collection, the LIBS detector software and hardware must be fully integrated into the existing laser ablation system. Applied Spectra is the only company that can offer such a LIBS system that can be easily integrated into the existing laser ablation system.

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7. Other Information.

This upgraded LIBS system is expected to improve the analytical capabilities of the PlasmaLab by providing simultaneous elemental compositional data associated with Sr, Hf, Nd and Pb isotopic measurements currently performed. The current LIBS detector and configuration does not allow for useful elemental compositional data to be acquired with the optimum conditions for isotopic analysis and this upgraded LIBS detector and cell window will.

8. The Efforts to Identify Additional Sources Including the Market Research Conducted.

There are several companies that make LIBS spectrometers, however none of these can integrate the data collection and software control into the existing laser system. Another product could not be integrated into the existing instrumentation and software that would allow us to maintain continuity of instrumentation, software, data, operating procedures. Applied Spectra does not have any authorized resellers among whom competition can be sought.

9. Future Plans to Permit Competition.

Future requirements for same/similar needs will undergo market research and be competed to the maximum extent practical.

10. Recommendation and Certification from Program Office.

Based on the above, I recommend this acquisition be conducted on a sole source basis and certify the above statements are true and accurate.

[REDACTED]

[REDACTED]
Signature/Date

[REDACTED]

11. Certifications from the Contracting Officer:

- a. Price reasonableness will be based on Applied Spectra, Inc.'s current price list and paid invoices.
- b. This justification is accurate and complete to the best of my knowledge and belief.

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12. Approvals

[REDACTED] [REDACTED]

Lisa D. Williams, Contracting Officer/Date

[REDACTED] [REDACTED]

Jennifer Kelley, Bureau Competition Advocate/Date

The U.S. Geological Survey (USGS) considers employee signatures to be Personal Protected Information, and therefore all employee signatures have been redacted. By posting this document, the Contracting Officer attests all appropriate signatures have been obtained.